

Decimal, Binary & Hexadecimal

Decimal Base 10

- Number system we use every day. Contains 10 digits:
↳ 0 1 2 3 4 5 6 7 8 9

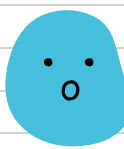
📌 123 in decimal means one hundred, two tens and three ones

Binary Base 2

- computers use this system every day. It has 2 digits:
↳ 0 1

📌 1010 in binary means 1 eight, 0 fours, 1 two and 0 ones.

But... why? Since Binary is a base-2 number system each place in a binary number represents a power of 2.
→ Also it's base-2 cause it only uses 2 digits: 0 & 1.

1	0	1	0	]	$2^0 = 1$	
2^3	2^2	2^1	2^0		$2^1 = 2$	
					$2^2 = 4$	
					$2^3 = 8$	
					... and so forth	

Hexadecimal Base 16

- This system is often used in computing to simplify binary. It has 16 digits: 0-9 and then A-F where A=10, B=11, ..., F=15).

📌 1A3 in a hexadecimal means 1 two hundred fifty six, A (or 10) sixteens, and 3 ones.

But... Why? Just like decimal (base-10) uses powers of 10 (like 1, 10, 100), hexadecimal uses powers of 16. Each place in a hexadecimal number represents a power of 16

1	A	3
16^2	16^1	16^0



Binary Conversion



ALI CAT
WE'RE IN THIS ADVENTURE TOGETHER!



Binary to Decimal

- 2^n ; Multiply each digit by 2 raised to the power of its position then add the results

Ex: 11011110 → $\begin{matrix} 1 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \\ 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \end{matrix}$] • We only add up the '1's

$$2^7 + 2^6 + 2^4 + 2^3 + 2^2 + 2^1 \\ \rightarrow 128 + 64 + 16 + 8 + 4 + 2 = 222$$

So the binary number 11011110 is equal to 222 in decimal.

Binary to Hexadecimal

- Group the binary digits into sets of 4, starting from the right
- Convert each group into its hexadecimal equivalent.

Ex: 00111100 → $\begin{matrix} \text{Group} & & \text{Group} \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 2^3 & 2^2 & 2^1 & 2^0 & 2^3 & 2^2 & 2^1 & 2^0 \end{matrix}$



Hexadecimal Table

Hexadecimal	Binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A (10)	1010
B (11)	1011
C (12)	1100
D (13)	1101
E (14)	1110
F (15)	1111

$$2^1 + 2^0 = 2 + 1 = 3$$

$$2^3 + 2^2 + 2^0 = 8 + 4 + 1 = 13$$

→ 13 in Hexadecimal is 'D'

Now that we have our two Hexadecimal digits we combine them

→ 00111100 is 3D
Binary → Hexadecimal



Decimal Conversion

Decimal to Binary



- Divide the decimal number by 2 and write down the remainder
↳ Repeat this until the quotient becomes 0.

Ex: 23

$$\begin{aligned} 23/2 &= 11 \text{ remainder } 1 \\ 11/2 &= 5 \text{ remainder } 1 \\ 5/2 &= 2 \text{ remainder } 1 \\ 2/2 &= 1 \text{ remainder } 0 \\ 1/2 &= 0 \text{ remainder } 1 \end{aligned}$$

$$\begin{array}{r} 11 \\ 2 \overline{) 23} \\ \underline{20} \\ 3 \\ 2 \\ \underline{2} \\ 1 \end{array} \quad \begin{array}{r} 5 \\ 2 \overline{) 11} \\ \underline{10} \\ 1 \end{array} \quad \begin{array}{r} 2 \\ 2 \overline{) 5} \\ \underline{4} \\ 1 \end{array} \quad \begin{array}{r} 1 \\ 2 \overline{) 2} \\ \underline{2} \\ 0 \end{array} \quad \begin{array}{r} 0 \\ 2 \overline{) 1} \\ \underline{0} \\ 1 \end{array}$$

- Write the remainders in reverse order, so from bottom to top we have: 10111

↳ So 23 (decimal) to binary is: 10111

Decimal to Hexadecimal

- Divide the decimal number by 16 and write down the remainder
→ Repeat this until the quotient becomes 0.

Ex: 345

$$\begin{aligned} 345/16 &= 21 \text{ remainder } 9 \\ 21/16 &= 1 \text{ remainder } 5 \\ 1/16 &= 0 \text{ remainder } 1 \end{aligned}$$

$$\begin{array}{r} 21 \\ 16 \overline{) 345} \\ \underline{32} \\ 25 \\ 16 \\ \underline{9} \end{array} \quad \begin{array}{r} 1 \\ 16 \overline{) 21} \\ \underline{16} \\ 5 \end{array} \quad \begin{array}{r} 0 \\ 16 \overline{) 1} \\ \underline{0} \\ 1 \end{array}$$

- Write the remainders in reverse order, so from bottom to top we have: 159

↳ So 345 (decimal) to Hexadecimal is: 159

Hexadecimal Conversions

Hexadecimal to Binary

- Break the hexadecimal number into its digits

Ex: 3 A F

3	A (10)	F (15)
0011	1010	1111
$2^3 2^2 2^1 2^0$	$2^3 2^2 2^1 2^0$	$2^3 2^2 2^1 2^0$
$2^1 + 2^0 = 3$	$2^3 + 2^1 = 10$	$2^3 + 2^2 + 2^1 + 2^0 = 15$



So: 3 A F (Hexadecimal)
 $\downarrow \quad \downarrow \quad \downarrow$
 0011 1010 1111 (Binary)

Hexadecimal to Decimal

- So Hexadecimal (base 16) uses powers of 16. Each place in a hexadecimal number represents a power of 16, starting from the right:

$$\begin{aligned} \rightarrow 16^0 &= 1 \\ 16^1 &= 16 \\ 16^2 &= 256 \\ \vdots \\ 16^n &= 16^n \end{aligned}$$



Ex: 2F

	hex	dec	
2 F	F is 15		$\rightarrow 2 \times 16^1 = 2 \times 16 = 32$ $15 \times 16^0 = 15 \times 1 = 15$
$16^1 \quad 16^0$	2 is 2		
			$\rightarrow 32 + 15 = 47$

So 2F in hexadecimal is 47 in decimal

